

Title:

Robust Short News Classification with Preprocessing and DistilBERT

Author:

Yide Fang

Project Type:

This project is an empirical NLP project. I will complete this project individually.

Problem Definition and Motivation:

This project studies short-text news topic classification. Given an input text sequence $x = (w_1, w_2, \dots, w_n)$, the goal is to predict a label y from a fixed set of news categories. I plan to use a four-class setting, such as World, Sports, Business, and Sci/Tech.

This task is important because text classification is widely used in content organization, recommendation systems, and information retrieval. It is also a practical NLP problem that can be studied with standard datasets and clear evaluation metrics. Short texts are especially challenging because they contain limited context, and small changes in wording, punctuation, or spelling may affect model performance. Because of this, I want to study not only overall classification accuracy, but also model robustness under noisy input.

The main question of this project is: Can simple preprocessing and lightweight transformer fine-tuning improve the robustness of short-text classification?

Project Outline

I plan to use the AG News dataset as the main benchmark for this project. This dataset is a standard text classification dataset and is well suited for a course project because it is clean, manageable, and widely used.

For software and implementation, I plan to use:

- Python
- PyTorch
- Hugging Face Transformers
- scikit-learn

Because this is an individual project, I want to keep the scope focused and realistic. My project will compare three levels of methods:

1. Simple baselines

I will first implement simple baseline methods, including:

Majority-class baseline
TF-IDF + Logistic Regression

These methods are easy to implement and will provide a strong point of comparison.

2. Pretrained transformer model

I will then fine-tune DistilBERT for the same classification task. I chose DistilBERT because it is smaller and faster than larger transformer models, which makes it more practical for an individual course project.

3. Robustness-oriented improvements

Finally, I will test a small number of simple improvements, such as:

- lowercasing
- punctuation normalization
- whitespace cleanup
- optional light data augmentation on part of the training set

The goal is to see whether these low-cost changes can make the model more robust when the input text is slightly noisy.

4. surface-level perturbations

In addition to surface-level perturbations, I will also explore a small semantic robustness extension based on adversarial examples. Inspired by instructor feedback, I will test whether the classifier can be misled by spurious category cues that do not change the main topic of the article. For example, I may inject short mentions of foreign countries, diplomats, or government officials into Sports, Business, or Sci/Tech articles in order to increase the likelihood of a World prediction without changing the original label. These adversarial examples can be generated with simple templates or with limited LLM assistance.

Experimental Design:

I will evaluate the models on three settings:

- Original clean test data
- A perturbed version of the test data with surface-level noise
- A small adversarial test subset with misleading semantic cues

The perturbed test set will be constructed by adding small realistic changes, such as:

- random capitalization changes
- punctuation removal

- extra spaces
- a limited number of synthetic typos

The adversarial subset will be constructed by adding short irrelevant but misleading category cues to selected examples while preserving the original label. For instance, I may add mentions of foreign countries or government officials to non-World articles in order to test whether the classifier relies on shallow correlations instead of true topic understanding.

This design allows me to compare not only standard performance and robustness to noisy input, but also robustness to targeted semantic perturbations.

The main evaluation metrics will be:

- Accuracy
- Macro-F1
- Per-class F1

I will compare all methods using the same train/test split. The simplest baseline will be the majority-class baseline, and the stronger traditional baseline will be TF-IDF + Logistic Regression. The main neural model will be fine-tuned DistilBERT, with and without the preprocessing improvements.

In addition to Accuracy, Macro-F1, and Per-class F1, I will compare performance drops from clean to noisy and from clean to adversarial settings in order to measure robustness more directly.

I will consider the project successful if:

- DistilBERT performs better than the simple baselines on clean data, and/or
- Preprocessing improves model performance on noisy data, even if the improvement on clean data is small, and/or
- The experiments reveal clear differences between models in their robustness to targeted adversarial cues.

In addition to reporting numerical results, I also plan to include a short error analysis. I will inspect examples that are classified correctly on clean input but incorrectly on noisy input, in order to better understand which kinds of perturbations are most harmful.

Expected Contribution:

This project will provide a focused comparison between traditional text classification methods and a lightweight pretrained transformer for short-text classification. More importantly, it will examine whether simple preprocessing can improve robustness in a realistic way, and whether the models are vulnerable to spurious semantic cues that do not change the true topic of the text. As an individual project, this topic remains appropriate because it uses standard tools, has a clear evaluation setup, and can be completed within the course timeline.

References:

Zhang, X., Zhao, J., and LeCun, Y. 2015. Character-level Convolutional Networks for Text Classification.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.

Sanh, V., Debut, L., Chaumond, J., and Wolf, T. 2019. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter.

Wei, J. and Zou, K. 2019. EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks.

Wolf, T. et al. 2020. Transformers: State-of-the-Art Natural Language Processing.